**Abstract**

Many influential perspectives on alignment assume that post-training is largely a process of re-organising mechanisms that a model learnt during pretraining. Is this true, and can it be observed directly? I intend to study this question in weight-space: given a decomposition of a model's weights into a collection of rank-one components (using VPD, a recently developed weight-based interpretability tool), to what extent do post-training weight updates lie within these components? If most weight updates are explained by parameter components, then this is evidence the model is re-using existing machinery, and also that it may be possible to continuously interpret a model during post-training without having to train a new decomposition at each stage. On the other hand, if weight updates typically fall outside of parameter components, then either post-training is creating new mechanisms, or the decomposition is blind to some structures of the model.

**Introduction**

Parameter Decomposition (https://www.goodfire.com/research/interpreting-lm-parameters) is an approach to mechanistic interpretability which decomposes the weights of a neural network into a sum of rank-one matrices called sub-components, plus some residual. These sub-components are optimised for various properties, most notably faithfully reconstructing a model's output on a given prompt with a minimal number of subcomponents and ensuring this reconstruction is robust against adversarial ablations on other subcomponents. This latter property is intended to ensure that the chosen subcomponents are causally important for a given task - in other words that they pinpoint the subcomponents sufficient for a given output.

In post-training, a model's weights are updated (using SFT or RL) to turn a model from a general-purpose base model into a useful chat assistant. It is widely believed that pre-training creates a host of mechanisms in a network which post-training re-weights, as opposed to post-training creating mechanisms of its own. This informal perspective - which I'll call the reweighting hypothesis - has been extremely influential in inspiring various perspectives on alignment. For example, the persona selection model (PSM, https://alignment.anthropic.com/2026/psm/ ) claims that "Pre-training teaches an LLM a distribution over personas … Post-training can be viewed as updating this distribution using training episodes as evidence". Steering vectors, alignment pre-training and explanations for emergent misalignment often rely on this hypothesis too. Existing evidence for re-weighting is typically behavioural or activation based. What remains unaddressed is how individual weight updates map onto existing mechanisms.

**Proposal**

My project intends to study whether post-training updates can be decomposed across VPD parameter components. This has immediate significance for the re-weighting hypothesis discussed in the previous paragraph. Assuming that parameter decomposition yields an accurate description of how a model completes some task, then if a post-training update factors

cleanly onto parameter subcomponents, this is evidence the update is re-weighting existing mechanisms. On the other hand, if post-training updates reliably lie outside the span of parameter subcomponents, then this is evidence against the re-weighting hypothesis (or that the parameter decomposition was incomplete).

If it turns out that post-training updates can be easily transformed into subcomponent updates, then it also gives the possibility for continually interpreting a model during post-training, rather than having to decompose the model from scratch at each step. This would be a powerful tool for monitoring post-training. For example, this may eventually be useful for distinguishing between an RL environment teaching a model the intended capability as opposed to teaching it to placate the grader. Note that this use case does not follow immediately from a successful outcome of this project because we still do not know what high-level behaviours like reward hacking look like as collections of subcomponents. Nevertheless, I believe it is a compelling motivation for exploring this line of research, and would constitute progress towards the second stage of Hubinger's interpretability tech tree (https://www.lesswrong.com/posts/nbq2bWLcYmSGup9aF).

**Preliminary results**

So far, I have run some basic tests on an existing decomposition of a relatively small 67M language model. Previous work taught this model to complete every emoticon with ":o" via LoRA fine-tuning and manually editing the parameter subcomponents. I have investigated various methods for mapping these fine-tuning updates onto the parameter decomposition. My main observation so far has been that the weight updates do not lie fully in the span of subcomponents, but that simply projecting the update onto subcomponents is sufficient for learning the task and even decreases off-target drift. This appears to very weakly support the existing-machinery hypothesis on this very basic task, although there is still a large portion of each update that remains unexplained by the subcomponents.

The next steps for this project are to understand these findings more deeply, before comparing the effects when teaching a model a new task, and ultimately scale it up to more interesting tasks on more realistic model sizes. I expect that any resulting method will break down as the number of updates increases so am curious to explore this trade-off. Finally, I am very interested in comparing the effects of fine-tuning versus reinforcement learning updates, since these are known to play different roles in post-training (e.g. https://arxiv.org/abs/2501.17161).

**Risks**

There a number of risks I wish to acknowledge about this project:

- It may turn out that post-training updates cannot be mapped onto subcomponents because parameter decomposition misses out on some important structure of model weights. This type of negative result would be of limited interest to the questions I am hoping to address. However, I think it could still be of some value to the parameter decomposition community.

- If this project is able to improve our understanding of what post-training is doing, it may be used to develop better parameter-efficient fine-tuning techniques and thereby accelerate capabilities research. This type of dual-use risk is typical for much of interpretability research. My personal view is that such research is still worth doing, so long as it is differentially more useful to safety than capabilities research. Whether that is true for my project will depend on what the results are, and so this tradeoff is something I will keep in mind.
- Relatedly, actively training against this method risks becoming an instance of "The Most Forbidden Technique" (https://www.lesswrong.com/posts/mpmsK8KKysgSKDm2T/). I do not endorse training against this method.
- A final risk of this project is that I will be unable to empirically study the structure of RL updates because there are no parameter decompositions for models of sufficient size to be worth training with RL. If this is the case, then the project will be constrained to studying the effects of fine-tuning, thereby missing out on a large (and possibly more significant) part of how models are post-trained in the real world.

1000 words (excluding these ones)

Another way of framing this project is trying to adapt the existing literature on model diffing (https://www.lesswrong.com/w/model-diffing) into parameter space. Existing work on model diffing has largely focused on how models differ in activation space (e.g. by comparing SAE features, as in https://transformer-circuits.pub/2024/model-diffing/index.html). One potential advantage of diffing in parameter space is that a training update is nothing but a modification of parameters, and so interpreting this modification directly seems to be more efficient than mediating through activations.